

CERA Medical Developer 02 Commerce Data and Integrations

Ten day execution manual with feature ownership and Git commands

Use this manual together with the CERA Medical Product Requirements Document

Document	Value
Owner	Stack & Scale for CERA Medical
Version	1.0
Date	17 September 2026
Status	Approved implementation baseline subject to CERA content and release approval

Deliver the data model, service catalogue, enquiry and operations APIs, customer ownership rules, Zoho synchronization, and Resend email delivery.

1 Role boundaries

Item	Working rule
Primary mission	Deliver the data model, service catalogue, enquiry and operations APIs, customer ownership rules, Zoho synchronization, and Resend email delivery.
Review relationship	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.
Integration branch	develop. It deploys to staging after required checks.
Production branch	main. Only approved release or hotfix pull requests may merge.
Daily evidence	Pushed branch, updated issue, tests, PR or staging link, and explicit next handoff.
Non-negotiable	Never commit secrets or production data; never push directly to develop or main; never disable authorization, tests, backups, or release controls to meet schedule.

2 Owned feature catalogue

ID	Feature	Acceptance evidence	Day	Dependencies
FND-003	Shared contracts and fixtures	Web, API, and tests import the same contracts; contract tests reject incompatible changes.	Day 1	FND-001
CAT-201	Vendure service catalogue	Seeded services are available through a stable read API and unpublished services never appear publicly.	Day 2	FND-003
ENQ-401	Secure enquiry submission	Valid enquiries are stored once; invalid requests return field-level errors; duplicate submissions do not create duplicate records.	Day 3	FND-003, CAT-201
ENQ-403	Enquiry status model	Invalid transitions are rejected; customers never receive internal notes or staff-only status values.	Day 6-7	ENQ-401
CUS-601	Customer account and enquiry claiming	One account cannot claim another email address's enquiry; expired or reused claim tokens fail safely.	Day 6	AUTH-501, ENQ-401

ID	Feature	Acceptance evidence	Day	Dependencies
OPS-701	Operations enquiry queue	Operations users can process seeded enquiries without database access; unauthorized roles receive no data.	Day 7	AUTH-502, ENQ-403
OPS-702	Internal notes and audit history	Customer APIs and exports contain no internal notes; audit events show who changed what and when.	Day 7	OPS-701
INT-801	Zoho CRM synchronization	A retry updates the same CRM record; failures enter a retry queue and remain visible without losing the enquiry.	Day 4	ENQ-401
INT-802	Resend transactional email	A repeated job sends one email; verified webhook events update delivery status; bounce and complaint events are recorded.	Day 5	ENQ-401
QA-1101	Automated release test suite	Required checks pass from a clean clone; failures are reproducible locally; test fixtures contain no real customer data.	Day 8	All build features
REL-1101	CERA user acceptance testing	Every launch-blocking defect is closed; lower-priority defects have an owner and accepted follow-up date; approvals are recorded.	Day 9	QA-1101
DOC-1201	Operations handover	A named CERA owner can execute routine content and enquiry tasks without developer assistance.	Day 10	REL-1102

3 Daily Git routine

3.1 Start every morning

First protect any unfinished local work. Then update develop and merge it into the active feature branch. This project does not require rebasing shared branches, so developers do not need force push.

```
git status
# If needed: git stash push -u -m "WIP before morning sync"
git fetch --prune origin
git switch develop
git pull --ff-only origin develop
git switch <your-feature-branch>
git merge --no-edit develop
# If used: git stash pop
pnpm install --frozen-lockfile
pnpm lint
pnpm typecheck
pnpm test
```

3.2 Start the assigned branch

```
git fetch --prune origin
git switch develop
git pull --ff-only origin develop
git switch -c <branch-from-the-daily-plan>
```

3.3 Commit and push

```
git status
git diff
git add -p
git diff --cached
git commit -m "<type>(<scope>): <specific result>"
git push -u origin $(git branch --show-current)
```

3.4 Open the pull request

```
gh pr create \
  --base develop \
  --head $(git branch --show-current) \
  --title "<type>(<scope>): <specific result>" \
  --body-file .github/pull_request_template.md
gh pr checks --watch
gh pr view --web
```

3.5 Resolve conflicts

```
git status
# Edit conflicted files and remove every conflict marker.
git diff --check
git add <resolved-files>
git commit
pnpm lint && pnpm typecheck && pnpm test
git push
# To cancel before commit: git merge --abort
```

3.6 Review and cleanup

```
gh pr checkout <PR_NUMBER>
pnpm install --frozen-lockfile
pnpm lint && pnpm typecheck && pnpm test && pnpm build
gh pr review <PR_NUMBER> --approve
# Maintainer after approval and green checks:
gh pr merge <PR_NUMBER> --squash --delete-branch
git switch develop
git pull --ff-only origin develop
git branch -d <merged-feature-branch>
git fetch --prune origin
```

4 Ten day feature plan

Day 1

Branch	Feature IDs	Reviewer
chore/cera-201-contracts-data	FND-003	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Define packages/contracts, PostgreSQL schemas, migrations, seed fixtures, status enums, error envelopes, and API ownership rules.

Done evidence. Contracts compile, fixtures support Dev 1 mocks, and migrations run up and down against a clean local database.

Required handoff. Publish contracts and fixtures before Dev 1 depends on them.

Day 2

Branch	Feature IDs	Reviewer
feat/cera-202-service-catalogue	CAT-201	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Configure Vendure service categories, service records, slugs, display pricing, availability text, media references, and read API. Disable checkout routes.

Done evidence. Seed catalogue is queryable and only publishable services are returned.

Required handoff. Give Dev 1 seed IDs and slugs; give Dev 3 runtime commands.

Day 3

Branch	Feature IDs	Reviewer
feat/cera-203-enquiry-api	ENQ-401	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Build enquiry validation, consent timestamp, reference generation, rate limiting, duplicate protection, persistence, and audit event creation.

Done evidence. Valid, invalid, repeated, and rate-limited submissions pass automated integration tests.

Required handoff. Publish request, response, and error examples to Dev 1.

Day 4

Branch	Feature IDs	Reviewer
feat/cera-204-zoho-sync	INT-801	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Implement least-privilege Zoho OAuth, external enquiry ID mapping, Lead or Contact upsert, retry policy, dead-letter visibility, and reconciliation command.

Done evidence. Repeated test jobs update one Zoho record and a simulated outage retries without losing the enquiry.

Required handoff. Dev 3 stores OAuth values; Dev 1 verifies visible outcome.

Day 5

Branch	Feature IDs	Reviewer
feat/cera-205-resend-notifications	INT-802	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Send customer and staff emails with idempotency keys; verify delivery webhooks; record sent, delivered, bounced, and complained events.

Done evidence. Retries send one message, invalid signatures fail, and delivery status is visible to operations.

Required handoff. Dev 1 verifies email UX; Dev 3 configures webhook secret.

Day 6

Branch	Feature IDs	Reviewer
feat/cera-206-customer-ownership	CUS-601, ENQ-403	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Implement customer-safe status mapping, enquiry claim tokens, ownership checks, profile APIs, and status detail APIs.

Done evidence. Cross-account access tests fail closed and expired or reused claim links cannot attach enquiries.

Required handoff. Give Dev 1 customer APIs and Dev 3 authorization claims.

Day 7

Branch	Feature IDs	Reviewer
feat/cera-207-operations-api	OPS-701, OPS-702	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Implement staff queue filters, assignment, valid transitions, internal notes, audit history, and safe customer/CRM projections.

Done evidence. Operations users complete the seeded workflow and no private note leaves staff-only APIs.

Required handoff. Dev 3 validates roles; confirm staff labels with CERA.

Day 8

Branch	Feature IDs	Reviewer
test/cera-208-integration-suite	QA-1101	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Complete migration, contract, authorization, Zoho retry, Resend webhook, backup-restore data, and queue recovery tests.

Done evidence. CI passes from a clean database and all integration evidence is linked to the release issue.

Required handoff. Run restore and failure exercises with Dev 3.

Day 9

Branch	Feature IDs	Reviewer
fix/cera-209-uat-integrations	REL-1101	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Resolve UAT data and integration defects, reconcile staging records, and verify migration backward compatibility.

Done evidence. Staging contains clean test data and all launch-blocking integration defects are closed.

Required handoff. Do not change contracts without notifying Dev 1.

Day 10

Branch	Feature IDs	Reviewer
release/cera-210-api-v1	DOC-1201	Dev 1 reviews consumer contracts and user-visible labels. Dev 3 reviews migrations, secrets, authorization, queues, and recovery behavior.

Build work. Run production API smoke tests, verify worker queues, Zoho and Resend delivery, and hand over field, status, and recovery references.

Done evidence. Production verification and operations data guide are attached to the release record.

Required handoff. Hand over field mapping, statuses, and reconciliation commands.

5 Daily completion checklist

- Issue is assigned, in the correct status, and has current acceptance criteria.
- Branch name matches this manual and starts from the latest develop.
- Changes are focused, self-reviewed, and contain no secret or real customer data.
- Relevant lint, type check, unit, integration, browser, build, migration, or infrastructure checks pass.
- PR links the issue and includes testing, screenshots where relevant, environment changes, migration impact, and rollback behavior.
- Reviewer feedback is resolved and required approvals apply to the latest push.
- Merged behavior is verified in staging and evidence is linked to the issue.
- The next developer receives an explicit contract, URL, fixture, command, or blocker before the day ends.

6 Escalation and schedule protection

- Raise a blocker in the 09:30 stand-up and in the GitHub Issue. Do not wait until the end of the day.
- If an external service is unavailable, use the agreed mock adapter and keep the real integration issue open with an owner.
- If a task is too large, split a complete vertical slice into a new PR. Keep incomplete behavior disabled.
- After Day 8, accept only UAT defects, security fixes, release documentation, and changes required to launch safely.
- A security, authorization, data-loss, migration, backup, or rollback blocker stops release even if the calendar target is at risk.

7 Assignment confirmation

Named developer	Date assigned	Developer acknowledgement	Technical lead